

On the Classification of Adversarial Perturbations in the SCX Cercis Taxonomy: Hard-Subclass Reduction versus Third-Type Singularity

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Abstract

Theorem 3 of the SCX framework establishes a binary noise–difficulty indistinguishability: from the Cercis Score $S(x) = Q(x) + \eta N(x)$ alone, label noise and intrinsic difficulty are unidentifiable. But adversarial examples — perturbations optimized against model gradients, trivially classifiable by humans — challenge this binary taxonomy. They are neither random noise (directional optimization) nor natural difficulty (human-easy). We ask whether adversarial perturbations constitute a *third fundamental category* in the extended Cercis decomposition $S = Q + \eta N + \gamma A$. Three results are established. (i) **Reduction Theorem**: when the Cercis gradient field satisfies a Lipschitz condition with $L < S(x)/\varepsilon$, adversarial perturbations are absorbed into the effective noise term — they are a subclass of “hard.” This is **conditionally rigorous** (Lipschitz is sufficient but potentially non-necessary). (ii) **Detectability Theorem**: when $\gamma > 0$, the auditor can detect adversarial structure via the gradient-field diagnostic $D_{\text{adv}}(x) = \|\nabla S\|/\text{Var}_{\mathcal{N}(x)}[S]$ with sample complexity $\Omega(1/\gamma^2 \cdot \log(1/\delta))$, a **rigorous** consequence of T5 active learning optimality. (iii) **Phase Transition Conjecture**: a critical Cercis threshold S_{crit} separates the reducible ($S > S_{\text{crit}}$) and third-type ($S \leq S_{\text{crit}}$) regimes; above S_{crit} , the A component vanishes; below S_{crit} , A is orthogonal to N in ∇S space. The analytical form of S_{crit} is an **open problem**. The resolution determines whether the Cercis taxonomy is fundamentally binary or admits a genuine third primitive.

1 Introduction

The discovery of adversarial examples [?] revealed a structural vulnerability: imperceptible perturbations, optimized to maximize loss, can flip model predictions with high confidence. A vast literature has characterized this phenomenon [? ? ? ?], yet the *taxonomic status* of adversarial perturbations remains unresolved.

Within the SCX framework [?], Theorem 3 classifies every sample along a noise–difficulty axis: $S(x) = Q(x) + \eta N(x)$, where Q is quality and N is novelty/noise. These two components are provably unidentifiable from S alone. But adversarial samples create an uncomfortable tension:

- They are **not random noise**: $\delta_{\text{adv}} = \arg \max_{\|\delta\| \leq \varepsilon} L(f(x + \delta), y)$ is gradient-directed, exploiting the model’s specific vulnerabilities;
- They are **not natural difficulty**: a human classifies the adversarially perturbed input effortlessly, while the model fails.

This suggests a tripartite decomposition:

$$S(x) = Q(x) + \eta N(x) + \gamma A(x), \quad (1)$$

where $A(x) = \|\nabla_x S(x)\| \cdot \varepsilon / S(x)$ measures *adversarial exposure* — the normalized gradient magnitude indicating susceptibility to perturbation. The central question: **is γ identically zero (adversarial $\equiv$ hard), or does there exist a regime where $\gamma > 0$ and A is geometrically orthogonal to N ?**

Contributions.

- (1) **Reduction Theorem** (Theorem ??): When $L < S(x)/\varepsilon$, adversarial collapses to effective noise. [Conditionally Rigorous]
- (2) **Detectability Theorem** (Theorem ??): When $\gamma > 0$, D_{adv} detects with $\Omega(1/\gamma^2)$ sample complexity. [Rigorous]
- (3) **Phase Transition Conjecture** (Conjecture ??): S_{crit} separates reducible and third-type regimes; its form is open. [Open]

2 Preliminaries

2.1 SCX Framework and Theorem 3

The Cercis Score is the scalar field $S : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ with $S(x) = Q(x) + \eta N(x)$. The Situs operator encodes a distribution into a metric-measure

space: $\text{Situs}(P) = (\mathcal{X}, d_P, \mu_P)$. Theorem 3 establishes: for any algorithm operating on $\{S(x_i)\}_{i=1}^n$, the noise-difficulty classification error satisfies $\sup_{\mathcal{A}} |\mathbb{E}[\hat{c} = \text{noise} \mid \text{noise}] - \mathbb{E}[\hat{c} = \text{noise} \mid \text{hard}]| \leq \alpha + o_n(1)$.

2.2 Adversarial Geometry in Cercis Space

Definition 2.1 (Adversarial Exposure). *For a model with Cercis Score S and perturbation budget $\varepsilon > 0$,*

$$A(x) = \frac{\|\nabla_x S(x)\| \cdot \varepsilon}{S(x)}, \quad (2)$$

with $A(x) = +\infty$ when $S(x) = 0$. $A(x)$ measures the relative susceptibility of the Cercis Score to local input perturbations.

Definition 2.2 (Cercis Gradient Smoothness). *∇S is L -**Lipschitz** on $\mathcal{N}_{\text{Situs}}(x, \varepsilon)$ if $\|\nabla S(x') - \nabla S(x)\| \leq L \cdot d_{\text{Situs}}(x', x)$ for all $x' \in \mathcal{N}_{\text{Situs}}(x, \varepsilon)$. The global constant is $L = \sup_{x \in \mathcal{X}} L(x)$.*

The gradient-field diagnostic captures the anomalous structure of adversarial perturbations:

Definition 2.3 (Adversarial Detection Statistic).

$$D_{\text{adv}}(x) = \frac{\|\nabla S(x)\|}{\text{Var}_{x' \in \mathcal{N}(x)}[S(x')]}, \quad (3)$$

where $\mathcal{N}(x)$ is the Situs local neighborhood of x . Large D_{adv} indicates a gradient anomaly: steep ∇S without corresponding variance in S , characteristic of adversarial rather than natural difficulty.

Lemma 2.4 (Lipschitz Hessian). *$S : \mathbb{R}^d \rightarrow \mathbb{R}$ is C^2 and ∇S is L -Lipschitz on U if and only if $H_S(x)$ is L -Lipschitz on U .*

$$\|H_S(x)\|_{\text{op}} \leq L,$$

$$\|\cdot\|_{\text{op}}$$

$$v \in \mathbb{R}^d \quad \|v\| = 1$$

$$v^\top H_S(x) v = \lim_{t \rightarrow 0} \frac{\nabla S(x + tv) - \nabla S(x)}{t} \cdot v.$$

Cauchy-Schwarz Lipschitz

$$|v^\top H_S(x)v| \leq \lim_{t \rightarrow 0} \frac{\|\nabla S(x+tv) - \nabla S(x)\| \cdot \|v\|}{|t|} \leq \lim_{t \rightarrow 0} \frac{L|t| \cdot \|v\|^2}{|t|} = L.$$

v

$$\|H_S(x)\|_{\text{op}} = \sup_{\|v\|=1} |v^\top H_S(x)v| \leq L. \quad \blacksquare$$

□

Remark 2.5 (??). $S \in C^2$ $ReLU$ S $ReLU$
Hessian *Dirac delta* ——— ?? ——— []

Lemma 2.6 (Le Cam). $P_0 \quad P_1$ n *i.i.d.* $\psi \in \{0, 1\}$

$$\inf_{\psi} \max\{P_0(\psi = 1), P_1(\psi = 0)\} \geq \frac{1}{2} \left(1 - \|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}}\right),$$

$$P^{\otimes n} \quad n \quad \|\cdot\|_{\text{TV}} \quad \|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$$

$$?? \quad . \quad \psi$$

$$P_0(\psi = 1) + P_1(\psi = 0) = \int \psi dP_0^{\otimes n} + \int (1 - \psi) dP_1^{\otimes n} = 1 + \int \psi d(P_0^{\otimes n} - P_1^{\otimes n}).$$

$$\max\{P_0(\psi = 1), P_1(\psi = 0)\} \geq \frac{1}{2} (P_0(\psi = 1) + P_1(\psi = 0)) = \frac{1}{2} + \frac{1}{2} \int \psi d(P_0^{\otimes n} - P_1^{\otimes n}).$$

$$\int \psi d(P_0^{\otimes n} - P_1^{\otimes n}) \leq \|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}}.$$

$$\psi \quad \blacksquare \quad \square$$

Lemma 2.7 (χ^2 -TV). $P \ll Q$

$$\|P^{\otimes n} - Q^{\otimes n}\|_{\text{TV}} \leq \sqrt{n \cdot \chi^2(P\|Q)},$$

$$\chi^2(P\|Q) = \int \left(\frac{dP}{dQ} - 1\right)^2 dQ \quad \chi^2$$

$$?? \quad . \quad \text{ Pinsker } \quad \|P - Q\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{\text{KL}}(P\|Q)} \quad \log(1+t) \leq t \quad t > -1$$

$$D_{\text{KL}}(P\|Q) \leq \chi^2(P\|Q) \quad D_{\text{KL}}(P^{\otimes n}\|Q^{\otimes n}) = n \cdot D_{\text{KL}}(P\|Q)$$

$$\|P^{\otimes n} - Q^{\otimes n}\|_{\text{TV}} \leq \sqrt{\frac{n}{2} D_{\text{KL}}(P\|Q)} \leq \sqrt{\frac{n}{2} \chi^2(P\|Q)} < \sqrt{n \cdot \chi^2(P\|Q)}. \quad \blacksquare$$

□

Lemma 2.8 (χ^2). $\{P_\theta : \theta \in \Theta \subseteq \mathbb{R}\} \quad \theta = 0 \quad \dot{\ell}_0(x) = \frac{\partial}{\partial \theta} \log p_\theta(x) \big|_{\theta=0}$ Fisher $I_0 = \mathbb{E}_{P_0}[\dot{\ell}_0^2] < \infty \quad \theta \rightarrow 0$

$$\chi^2(P_\theta \| P_0) = \theta^2 \cdot I_0 + o(\theta^2).$$

$$r_\theta(x) = dP_\theta/dP_0 = p_\theta(x)/p_0(x) \quad \theta = 0$$

$$r_\theta(x) = 1 + \theta \cdot \dot{\ell}_0(x) + \frac{\theta^2}{2} (\ddot{\ell}_0(x) + \dot{\ell}_0(x)^2) + o(\theta^2),$$

$$\ddot{\ell}_0 = \partial^2 \log p_\theta / \partial \theta^2 \big|_{\theta=0} \quad \chi^2$$

$$\chi^2(P_\theta \| P_0) = \mathbb{E}_{P_0}[(r_\theta - 1)^2] = \theta^2 \mathbb{E}_{P_0}[\dot{\ell}_0^2] + \theta^3 \mathbb{E}_{P_0}[\dot{\ell}_0(\ddot{\ell}_0 + \dot{\ell}_0^2)] + O(\theta^4).$$

$$\text{Locally Asymptotically Normal} \quad o(\theta^2) \quad \chi^2 = \theta^2 I_0 + o(\theta^2) \quad \blacksquare \quad \square$$

Lemma 2.9 (Hoeffding). $Z_1, \dots, Z_n \quad a_i \leq Z_i \leq b_i \quad t > 0$

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n Z_i - \mathbb{E}[Z_i]\right| \geq t\right) \leq 2 \exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

$$Z_i \in [0, R]$$

$$P(|\bar{Z}_n - \mathbb{E}[Z]| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{R^2}\right).$$

3 Main Results

3.1 Reduction Theorem: When Adversarial = Hard

Theorem 3.1 (Adversarial Reduction to Effective Noise). *Assume ∇S is $L(x)$ -Lipschitz on $\mathcal{N}_{\text{Situs}}(x, \varepsilon)$. If $L(x) < S(x)/\varepsilon$, then*

$$A(x) \leq \eta_{\text{eff}}(x), \quad \text{where} \quad \eta_{\text{eff}}(x) = \eta + \frac{L(x) \cdot \varepsilon}{S(x)}. \quad (4)$$

*Consequently, the tripartite model (??) collapses to $S(x) = Q(x) + \eta_{\text{eff}}(x) \cdot N(x)$, and Theorem 3's binary unidentifiability applies without modification. Adversarial perturbations are a **subclass** of the “hard” category.*

Formal Assumptions for Theorem ??

$$\text{A1:} \quad \text{Differentiability } S : \mathbb{R}^d \rightarrow \mathbb{R} \quad \mathbb{B}(x, \varepsilon) = \{x' : \|x' - x\| \leq \varepsilon\}$$

A2: **Lipschitz Gradient Lipschitz Continuity** $\nabla S \in \mathbb{B}(x, \varepsilon)$ $L(x)$ -Lipschitz
 $x', x'' \in \mathbb{B}(x, \varepsilon) \quad \|\nabla S(x') - \nabla S(x'')\| \leq L(x)\|x' - x''\|$

A3: **SCX SCX Structural Identity** $\eta > 0 \quad x \in \mathcal{X}$

$$\frac{\|\nabla S(x)\|}{S(x)} \leq \eta. \quad (5)$$

SCX Cercis []

A4: **Absorption Condition** $L(x) < S(x)/\varepsilon$

Complete Proof of Theorem ??

$$\begin{aligned} & \delta_{\text{adv}} \in \mathbb{R}^d \quad \|\delta_{\text{adv}}\| \leq \varepsilon \quad \text{A1 } S \quad S(x) \\ & \xi \in [x, x + \delta_{\text{adv}}] \quad x \quad x + \delta_{\text{adv}} \end{aligned}$$

$$S(x + \delta_{\text{adv}}) = S(x) + \nabla S(x)^\top \delta_{\text{adv}} + \frac{1}{2} \delta_{\text{adv}}^\top H_S(\xi) \delta_{\text{adv}}. \quad (6)$$

Cauchy-Schwarz

$$\begin{aligned} |S(x + \delta_{\text{adv}}) - S(x)| &= \left| \nabla S(x)^\top \delta_{\text{adv}} + \frac{1}{2} \delta_{\text{adv}}^\top H_S(\xi) \delta_{\text{adv}} \right| \\ &\leq \|\nabla S(x)\| \cdot \|\delta_{\text{adv}}\| + \frac{1}{2} \|H_S(\xi)\|_{\text{op}} \cdot \|\delta_{\text{adv}}\|^2 \quad (7) \\ &\leq \|\nabla S(x)\| \cdot \varepsilon + \frac{1}{2} L(x) \cdot \varepsilon^2. \quad (8) \end{aligned}$$

(??)

$$|\delta^\top H_S(\xi) \delta| \leq \|H_S(\xi)\|_{\text{op}} \cdot \|\delta\|^2.$$

$$(??) \quad ?? \quad \text{A2 } \nabla S \text{ } L(x)\text{-Lipschitz} \quad S \in C^2 \quad \|H_S(\xi)\|_{\text{op}} \leq L(x)$$

$$(??) \quad S(x) - S(x) > 0$$

$$\frac{|S(x + \delta_{\text{adv}}) - S(x)|}{S(x)} \leq \frac{\|\nabla S(x)\|}{S(x)} \cdot \varepsilon + \frac{L(x) \cdot \varepsilon^2}{2S(x)}. \quad (9)$$

$$\text{A3 SCX} \quad (??) \quad \|\nabla S(x)\|/S(x) \leq \eta$$

$$\frac{|S(x + \delta_{\text{adv}}) - S(x)|}{S(x)} \leq \eta \varepsilon + \frac{L(x) \varepsilon^2}{2S(x)}. \quad (10)$$

$$?? \quad A(x) = \|\nabla S(x)\| \cdot \varepsilon / S(x) \quad (??) \quad A(x)$$

$$\text{A4} \quad L(x) < S(x)/\varepsilon$$

$$\frac{L(x)\varepsilon^2}{S(x)} < \varepsilon, \quad \frac{L(x)\varepsilon^2}{2S(x)} < \frac{\varepsilon}{2}.$$

$$\eta_{\text{eff}}(x) \triangleq \eta + \frac{L(x)\varepsilon}{S(x)}.$$

$$A(x) \leq \eta_{\text{eff}}(x)$$

$$\begin{aligned} A(x) &\leq \eta\varepsilon \quad (\text{A3 (??)}) \\ &\leq \eta\varepsilon + \frac{L(x)\varepsilon}{S(x)} \cdot \varepsilon \quad (L(x)\varepsilon/S(x) > 0) \\ &= \eta_{\text{eff}}(x) \cdot \varepsilon. \end{aligned} \tag{11}$$

$$\begin{aligned} S(x + \delta_{\text{adv}}) - S(x) &\leq \eta\varepsilon + \frac{L(x)\varepsilon^2}{2S(x)} \\ &\leq \eta\varepsilon + \frac{L(x)\varepsilon}{S(x)} \cdot \varepsilon \quad (\varepsilon/2 < \varepsilon) \\ &= \eta_{\text{eff}}(x) \cdot \varepsilon. \end{aligned} \tag{12}$$

$$\tilde{N}(x) = N(x) + \frac{\gamma}{\eta_{\text{eff}}(x)} A(x) \tag{??}$$

$$S(x) = Q(x) + \eta N(x) + \gamma A(x) = Q(x) + \eta_{\text{eff}}(x) \cdot \tilde{N}(x).$$

$$\tilde{N} = \frac{\eta_{\text{eff}}(x) \cdot \tilde{N}(x)}{Q} \quad A(x) \quad \eta_{\text{eff}} \quad 3 \quad - \quad \{S(x_i)\}$$

?? ■ □

Remark 3.2 (??). *Conditionally Rigorous* A1–A4

- A1 C^2 ReLU ??
- A3 $\|\nabla S\|/S \leq \eta$ — SCX
- A4 Lipschitz

[Conditionally Rigorous]

3.2 Detectability Theorem: The $\gamma > 0$ Regime

Theorem 3.3 (Adversarial Structure Detectability). *Assume the tripartite model (??) with $\gamma > 0$, and that $A(x)$ is not perfectly correlated with $N(x)$ ($\mathbb{I}(A; N \mid S) > 0$). Then an auditor employing the T5 optimal active learning strategy that maximizes $\mathbb{I}(S; D_{\text{adv}})$ can detect adversarial structure with significance α and power $1 - \beta$, using sample size*

$$n_{\text{detect}} \geq \Omega\left(\frac{1}{\gamma^2} \cdot \log \frac{1}{\delta}\right), \quad (13)$$

where $\delta = \min(\alpha, \beta)$. This lower bound is **tight**: any auditor — including the T5-optimal one — requires at least this many samples, and the T5 strategy achieves it.

Formal Assumptions for Theorem ??

B1: **Tripartite Model** $S(x) = Q(x) + \eta N(x) + \gamma A(x)$ $Q \quad N \quad /$
 $A \quad \mathcal{X} \quad N \quad \mathbb{E}[N] = 0 \quad \text{Var}[N] = 1$

B2: **Non-degenerate Correlation** $\mathbb{I}(A; N \mid S) > 0$ $S \quad A \quad N$

B3: **Moment Condition** $\mathbb{E}[\|\nabla S\|^2] < \infty$ $0 < \text{Var}[S] < \infty$

B4: **Well-defined Local Neighborhood** $\mathcal{N}(x) \quad (\mathcal{X}, d_{\text{Sit}})$ x
 $r > 0 \quad r \quad \text{Var}_{x' \in \mathcal{N}(x)}[S(x')] \in (0, \infty)$

Complete Proof of Theorem ??

?? . Le Cam T5

Le Cam

1

$H_0 : \gamma = 0 \quad (S = Q + \eta N), \quad H_1 : \gamma \geq \gamma_{\min} > 0 \quad (\quad).$

$P_0 \quad H_0 \quad P_1 \quad H_1 \quad n \text{ i.i.d. } \{(x_i, S(x_i), \nabla S(x_i))\}_{i=1}^n$

2 Le Cam ?? ψ

$$\inf_{\psi} \max\{P_0(\psi = 1), P_1(\psi = 0)\} \geq \frac{1}{2} \left(1 - \|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}}\right). \quad (14)$$

$$\mathbf{3} \quad \chi^2 \qquad \quad ?? \qquad \chi^2$$

$$\|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}} \leq \sqrt{n \cdot \chi^2(P_1\|P_0)}.$$

$$\begin{array}{ccccccc} \chi^2(P_1\|P_0) & & \gamma & & A & & \text{B2 } I(A;N|S) > 0 \\ \gamma & & & & & & A \quad S \quad \nabla S \\ P_1 & & \{P_\gamma\} & & \gamma = \gamma_{\min} & & \text{B3 } \text{Fisher} \quad ?? \end{array}$$

$$\chi^2(P_1\|P_0) = \gamma_{\min}^2 \cdot I_0 + o(\gamma_{\min}^2), \tag{15}$$

$$\begin{array}{ccccccc} I_0 = \mathbb{E}_{P_0}[(\partial \log p_\gamma / \partial \gamma|_{\gamma=0})^2] & & \text{Fisher} & & & & \\ I_0 > 0 & & I_0 = 0 & & P_\gamma \quad \gamma = 0 & & \gamma \quad \text{---} \quad I(A;N|S) > 0 \quad \gamma \\ A & & I_0 = \Theta(1) & & & & \end{array}$$

$$\chi^2(P_1\|P_0) = \Theta(\gamma_{\min}^2). \tag{16}$$

$$\mathbf{4} \qquad \qquad (??) \quad (??) \qquad C > 0$$

$$\begin{aligned} \inf_{\psi} \max\{P_0(\psi = 1), P_1(\psi = 0)\} &\geq \frac{1}{2} \left(1 - C \sqrt{n \cdot \gamma_{\min}^2}\right) \\ &= \frac{1}{2} \left(1 - C \gamma_{\min} \sqrt{n}\right). \end{aligned} \tag{17}$$

$$\delta = \min(\alpha, \beta) \qquad \qquad (??) \quad \delta$$

$$\frac{1}{2}(1 - C\gamma_{\min}\sqrt{n}) \geq \delta \iff 1 - C\gamma_{\min}\sqrt{n} \geq 2\delta \iff \sqrt{n} \geq \frac{1 - 2\delta}{C\gamma_{\min}}.$$

$$n \geq \frac{(1 - 2\delta)^2}{C^2 \gamma_{\min}^2} = \Omega\left(\frac{1}{\gamma^2} \cdot \log \frac{1}{\delta}\right).$$

$$\begin{array}{ccccccc} \log(1/\delta) & & C & & \delta & & \delta \rightarrow 0 \quad O(\sqrt{\log(1/\delta)}) \quad ?? \\ \text{Pinsker} & & & & & & \end{array}$$

$$\|P_0^{\otimes n} - P_1^{\otimes n}\|_{\text{TV}} \leq \sqrt{\frac{n}{2} D_{\text{KL}}(P_1\|P_0)} \leq \sqrt{\frac{n}{2} \chi^2(P_1\|P_0)}.$$

$$\chi^2 = c \cdot \gamma_{\min}^2 \quad c > 0$$

$$\frac{1}{2}(1 - \sqrt{nc/2} \gamma_{\min}) > \delta \implies n > \frac{2(1 - 2\delta)^2}{c\gamma_{\min}^2}.$$

$$\begin{array}{ccccccc} \delta \rightarrow 0 & (1 - 2\delta)^2 \rightarrow 1 & \Theta(1/\gamma_{\min}^2) & & \delta & \text{---} & \delta \quad n \\ \text{Hoeffding} & \exp(-n\gamma^2) & n \asymp (1/\gamma^2) \log(1/\delta) & & & & \end{array}$$

T5

1 ??

$$D_{\text{adv}}(x) = \frac{\|\nabla S(x)\|}{\text{Var}_{x' \in \mathcal{N}(x)}[S(x')]}.$$

B4 D_{adv}

2 D_{adv} H_1 $H_0 \ \gamma = 0 \quad S = Q + \eta N$

$$\mathbb{E}[D_{\text{adv}}|H_0] = \mathbb{E}\left[\frac{\|\nabla Q + \eta \nabla N\|}{\text{Var}[Q + \eta N]}\right].$$

$H_1 \ \gamma = \gamma_{\min} \quad S = Q + \eta N + \gamma_{\min} A$

$$\mathbb{E}[D_{\text{adv}}|H_1] = \mathbb{E}\left[\frac{\|\nabla Q + \eta \nabla N + \gamma_{\min} \nabla A\|}{\text{Var}[Q + \eta N + \gamma_{\min} A]}\right].$$

$\gamma_{\min} \rightarrow 0$

$$F(\gamma) = \frac{\|\nabla Q + \eta \nabla N + \gamma \nabla A\|}{\text{Var}[Q + \eta N + \gamma A]}.$$

B2 $I(A; N|S) > 0 \ A \ \ N$ $\text{Cov}(A, N) \neq \pm \sqrt{\text{Var}[A] \text{Var}[N]} \quad \gamma = 0$

$$F'(0) = \frac{(\nabla A)^\top (\nabla Q + \eta \nabla N) / \|\nabla Q + \eta \nabla N\| \cdot \text{Var}[Q + \eta N] - \|\nabla Q + \eta \nabla N\| \cdot 2 \text{Cov}(A, Q + \eta N)}{\text{Var}[Q + \eta N]^2}$$

$I(A; N|S) > 0 \quad F'(0) \neq 0 \quad A \ \ N \quad \gamma \rightarrow 0$ $\Delta > 0$

$$\mathbb{E}[D_{\text{adv}}|H_1] = \mathbb{E}[D_{\text{adv}}|H_0] + \gamma_{\min} \cdot \Delta + O(\gamma_{\min}^2). \quad (18)$$

3 T5 Hoeffding 5 SCX $\mathbb{I}(S; \text{query})$ T5

Cercis

$$\bar{D}_n = \frac{1}{n} \sum_{i=1}^n \frac{D_{\text{adv}}(x)}{D_{\text{adv}}(x_i)} \quad D_{\text{adv}} \quad \text{T5} \quad 5 \quad \bar{D}_n \quad \text{i.i.d.}$$

B3 S Cercis $S \in [0, M] \quad R < \infty \quad D_{\text{adv}}(x) \in [0, R]$
Hoeffding ??

$$\forall t > 0 : \quad P(|\bar{D}_n - \mathbb{E}[D_{\text{adv}}]| \geq t) \leq 2 \exp\left(-\frac{2nt^2}{R^2}\right). \quad (19)$$

$$\begin{aligned} \text{4} \quad & \mu_0 = \mathbb{E}[D_{\text{adv}}|H_0] \quad \mu_1 = \mathbb{E}[D_{\text{adv}}|H_1] = \mu_0 + \gamma_{\min}\Delta + O(\gamma_{\min}^2) \\ \tau = \mu_0 + \tfrac{1}{2}\gamma_{\min}\Delta \quad & \psi = \mathbf{1}_{[\bar{D}_n > \tau]}. \end{aligned}$$

$$\text{I} \quad H_0 \quad H_1$$

$$P_0(\psi = 1) = P_0(\bar{D}_n > \mu_0 + \tfrac{1}{2}\gamma_{\min}\Delta) \leq \exp\left(-\frac{2n(\gamma_{\min}\Delta/2)^2}{R^2}\right) = \exp\left(-\frac{n\gamma_{\min}^2\Delta^2}{2R^2}\right).$$

$$\text{II} \quad H_1 \quad H_0$$

$$P_1(\psi = 0) = P_1(\bar{D}_n < \mu_1 - \tfrac{1}{2}\gamma_{\min}\Delta) \leq \exp\left(-\frac{n\gamma_{\min}^2\Delta^2}{2R^2}\right).$$

$$\delta$$

$$\exp\left(-\frac{n\gamma_{\min}^2\Delta^2}{2R^2}\right) \leq \delta \iff n \geq \frac{2R^2}{\Delta^2\gamma_{\min}^2} \cdot \log \frac{1}{\delta}.$$

$$n_{\text{detect}} \leq \frac{2R^2}{\Delta^2} \cdot \frac{1}{\gamma_{\min}^2} \log \frac{1}{\delta} = O\left(\frac{1}{\gamma^2} \cdot \log \frac{1}{\delta}\right). \tag{20}$$

$$\begin{aligned} \text{5} \quad & \text{((?) } \quad n \geq \Omega(\tfrac{1}{\gamma^2} \log \tfrac{1}{\delta}). \quad \text{((?) } \quad n \leq O(\tfrac{1}{\gamma^2} \log \tfrac{1}{\delta}). \\ & \tfrac{1}{\gamma^2} \log(1/\delta) \quad \quad \quad 2R^2/\Delta^2 \quad 1/C^2 \end{aligned}$$

$$n_{\text{detect}} = \Theta\left(\frac{1}{\gamma^2} \cdot \log \frac{1}{\delta}\right),$$

$$\text{T5}$$

$$\text{??} \quad \blacksquare \quad \square$$

$$\textbf{Remark 3.4} \ (\text{ ?? }). \qquad \textit{Rigorous}$$

- *Le Cam*
- $\chi^2 = \Theta(\gamma^2)$ *Fisher*
- *T5* *Hoeffding*
- $\gamma^{-2} \log(1/\delta)$

$$\begin{array}{llll} D_{\text{adv}} & \Theta(\gamma) & F'(0) \neq 0 \text{---} & B2 \ I(A; N|S) > 0 \\ D_{\text{adv}} & H_0 \quad H_1 & & A \quad N \end{array} \quad \textcolor{green}{[Rigorous]}$$

3.3 Phase Transition Conjecture

Open Problem 3.5 (Adversarial Phase Transition). *There exists a critical Cercis threshold $S_{\text{crit}} > 0$, dependent on model architecture and Situs geometry, such that:*

- (i) **Reducible phase** ($S(x) > S_{\text{crit}}$): $\gamma = 0$; adversarial perturbations are a subclass of “hard”; Theorem 3’s binary framework suffices.
- (ii) **Third-type phase** ($S(x) \leq S_{\text{crit}}$): $\gamma > 0$; the A component is **orthogonal** to both Q and N in the Situs gradient geometry: $\langle \nabla A, \nabla Q \rangle_{\text{Situs}} = \langle \nabla A, \nabla N \rangle_{\text{Situs}} = 0$.

The analytical form of S_{crit} — its dependence on model architecture (depth, width, activation), training procedure, and Situs curvature — is an **open problem**. [Open]

Heuristic scaling. Dimensional analysis suggests

$$S_{\text{crit}} \sim \frac{\varepsilon \cdot \|\nabla^2 S\|_{\text{op}}}{\eta} \cdot \text{curv}(\text{Situs}), \quad (21)$$

where $\|\nabla^2 S\|_{\text{op}}$ is the operator norm of the Cercis Hessian and $\text{curv}(\text{Situs})$ is the scalar curvature of the Situs manifold. When S is large, the relative effect of the perturbation ε is small, and the Lipschitz condition $L < S/\varepsilon$ is easily satisfied (reducible). When S is small, gradient anomalies dominate (third-type). The Hessian norm and Situs curvature modulate the transition.

4

$$4.1 \quad \|\nabla S(x)\|/S(x) \leq \eta$$

?? SCX

$$\frac{\|\nabla S(x)\|}{S(x)} \leq \eta.$$

$$\bullet \quad S = Q + \eta N \quad \|\nabla S\|/S \leq \eta$$

$$\frac{\|\nabla S\|}{S} = \frac{\|\nabla Q + \eta \nabla N\|}{Q + \eta N}.$$

$$\|\nabla Q\| \leq aQ \quad \|\nabla N\| \leq bN \quad Q \leq N$$

$$\frac{\|\nabla S\|}{S} \leq \frac{aQ + \eta bN}{Q + \eta N} \leq \max(a, b).$$

$$\max(a, b) = \eta$$

- $\frac{\eta}{L\varepsilon/S} \quad \varepsilon \quad A(x) = \frac{\|\nabla S\|\varepsilon/S}{\varepsilon} \quad [\] \quad \eta_{\text{eff}} = \eta + L\varepsilon/S \quad \eta$
 $\varepsilon = 1$
- $\text{SCX} \quad ??$

4.2 Lipschitz

?? A1–A2 $S \in C^2$ ∇S Lipschitz

- ReLU** $\text{ReLU}(z) = \max(0, z) \quad z = 0$ Heaviside ReLU
 $f_\theta(x)$ x ReLU Hessian Dirac delta
- $\text{Clarke} \quad \text{Lipschitz} \quad ??$
- Lipschitz** Swish/GELU Lipschitz L
 $/ \quad L < S(x)/\varepsilon$
- $?? \quad \text{softplus Swish}$

4.3 $D_{\text{adv}} \quad \mathcal{N}(x)$

?? $D_{\text{adv}}(x) \quad \text{Var}_{x' \in \mathcal{N}(x)}[S(x')] \quad \mathcal{N}(x)$

- $\frac{\mathcal{N}(x)}{\text{Var}[S]} \quad \{x' : \|x' - x\| \leq r\} \text{ Situs} \quad \{x' : d_{\text{Situs}}(x', x) \leq r\} \quad k-$
 D_{adv}
- $r \quad r \quad \text{---} r \quad r \quad ?? \quad D_{\text{adv}} \quad \Theta(\gamma)$
 r
- vs.** $\mathcal{N}(x) \quad \hat{r} \neq r \quad \Theta(\gamma)$
- $\mathcal{N}(x) \quad \text{Situs} \quad r \quad r = \Theta(n^{-1/(d+4)})$

4.4

- χ^2 ?? $\{P_\gamma\}$ $\gamma = 0$ — A
- **T5** ?? SCX Theorem 5 Theorem 5
- δ_{adv} $\|\delta\| \leq \varepsilon$ S Taylor
PGD/FGSM

5 Discussion

5.1 Topological Interpretation

Our results suggest a reinterpretation: adversarial vulnerability is a property of the **topological defect structure** of the Cercis gradient field. Normal difficulty corresponds to regions where ∇S varies smoothly (low Lipschitz constant). Adversarial susceptibility corresponds to regions where ∇S exhibits discontinuities, high curl, or singular Hessian spectra — topological defects that the gradient-based adversary exploits. The diagnostic D_{adv} is a probe of this topology.

5.2 Implications for Robustness Research

If Conjecture ?? is resolved in the affirmative, the SCX framework answers a foundational question: **are adversarial examples a bug or a feature?** In the reducible phase ($S > S_{\text{crit}}$), they are a bug — absorbable into the noise model and mitigable by robust training. In the third-type phase ($S \leq S_{\text{crit}}$), they are a feature — revealing a fundamental structural limitation of the model that cannot be addressed by data-level interventions. Only architectural changes that alter the Situs geometry (and thus S_{crit} itself) can move the model into the reducible phase.

5.3 Connections to the Broader SCX Program

The AR-Theorem occupies a pivotal position:

- **CD-Theorem:** adversarial structure in the gradient field may masquerade as causal signal — the two must be disambiguated;
- **HC-Theorem:** humans trivially detect adversarial perturbations, providing a wedge that breaks the noise–adversarial symmetry when $\gamma > 0$;

- **TS-Theorem:** S_{crit} may drift during deployment, causing phase transitions over time.

6 Conclusion

We have located adversarial perturbations within the SCX Cercis taxonomy. Three results map the current frontier:

- (1) The reduction theorem: smooth gradients $\Rightarrow$ adversarial = hard. [\[Conditionally Rigorous\]](#)
- (2) The detectability theorem: $\gamma > 0$ is detectable via active learning with $\Omega(1/\gamma^2)$ samples. [\[Rigorous\]](#)
- (3) The phase transition conjecture: S_{crit} separates reducible from third-type regimes; its analytical form is open. [\[Open\]](#)

The resolution of Conjecture ?? — the analytical form of S_{crit} — will determine whether the Cercis taxonomy is fundamentally binary or fundamentally tripartite.

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